do it, in our research. The bulk, for us, until the 1980s, was just a visual aid. An aid to give us intuition about what's going on in our mathematics, and to help us communicate with each other and with nonphysicists. A visual aid. Not a real thing.

What would it mean for the bulk to be real? How can we test whether it's real? The bulk is real if it can influence things we measure. And until the 1980s we saw no way it could influence our measurements.

Then in 1984 this changed. Radically. Michael Green at the University of London and John Schwarz at Caltech had a huge breakthrough in the quest to discover the laws of quantum gravity.[35] But strangely, their breakthrough worked only if our universe is a brane embedded in a bulk that has one time dimension and nine space dimensions—a bulk with six more space dimensions than our brane. In the mathematical formalism that Green and Schwarz were pursuing, called "superstring theory," the bulk's extra dimensions influence our brane in major ways, in ways that can be measured in physics experiments when we have sufficiently advanced technology. In ways that may make it possible to reconcile the laws of quantum physics with Einstein's relativistic laws.